

Презентация

Лабораторная работа №2

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Информация

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Освоить построение простых сетей на коммутаторе и маршрутизаторе, разобрать Ethernet, ARP и ICMP в Wireshark.

1. Собрать локальный сегмент из двух VPCS и Ethernet-коммутатора.
2. Настроить IP-адреса и проверить связь ping.
3. Включить захват пакетов на линии PC1 — коммутатор.
4. Разобрать ARP и ICMP в Wireshark.
5. Заменить второй VPCS на FRR.
6. Проверить доступность шлюза с PC1.
7. Заменить FRR на VyOS.
8. Сравнить поведение FRR и VyOS.

Два VPCS соединены через один Ethernet-коммутатор.

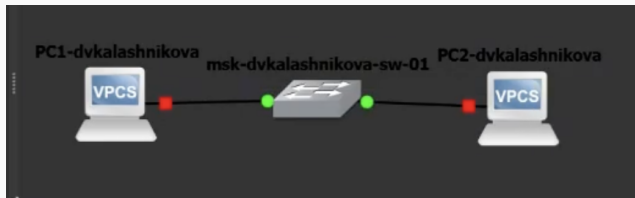
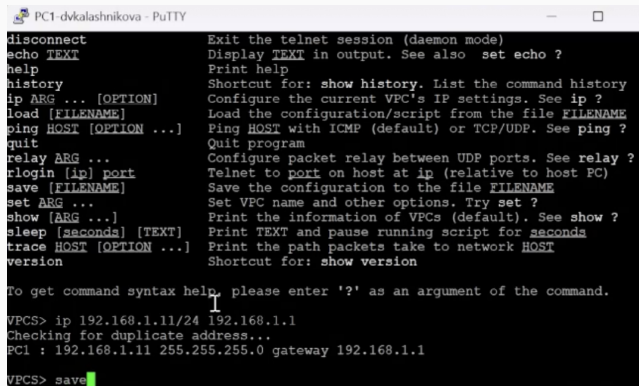


Рис. 1: Локальный сегмент в GNS3

Настройка PC1

Задан адрес 192.168.1.11/24, шлюз 192.168.1.1.



```
PC1-dvkalashnikova - PuTTY

disconnect      Exit the telnet session (daemon mode)
echo TEXT      Display TEXT in output. See also set echo ?
help           Print help
history        Shortcut for: show history. List the command history
ip ARG ... [OPTION] Configure the current VPC's IP settings. See ip ?
load [FILENAME] Load the configuration/script from the file FILENAME
ping HOST [OPTION ...] Ping HOST with ICMP (default) or TCP/UDP. See ping ?
quit           Quit program
relay ARG ...  Configure packet relay between UDP ports. See relay ?
rlogin [ip] port Telnet to port on host at ip (relative to host PC)
save [FILENAME] Save the configuration to the file FILENAME
set ARG ...    Set VPC name and other options. Try set ?
show [ARG ...] Print the information of VPCs (default). See show ?
sleep [seconds] [TEXT] Print TEXT and pause running script for seconds
trace HOST [OPTION ...] Print the path packets take to network HOST
version        Shortcut for: show version

To get command syntax help, please enter '?' as an argument of the command.

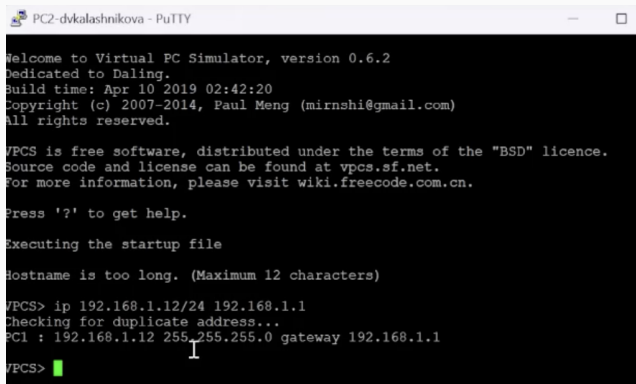
VPCS> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.11 255.255.255.0 gateway 192.168.1.1

VPCS> save
```

Рис. 2: Настройка PC1

Настройка PC2

Задан адрес 192.168.1.12/24 с тем же шлюзом.



```
PC2-dvkalashnikova - PuTTY

Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

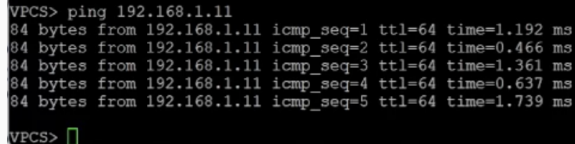
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0 gateway 192.168.1.1
VPCS> █
```

Рис. 3: Настройка PC2

Пять откликов с TTL 64 — коммутация работает.

A screenshot of a terminal window with a black background and white text. The prompt 'VPCS>' is followed by the command 'ping 192.168.1.11'. The output shows five successful ping responses, each with 84 bytes, a TTL of 64, and a response time in milliseconds. The response times are 1.192 ms, 0.466 ms, 1.361 ms, 0.637 ms, and 1.739 ms. The prompt 'VPCS>' is followed by a green cursor.

```
VPCS> ping 192.168.1.11
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.192 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=0.466 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.361 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=0.637 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.739 ms
VPCS> █
```

Рис. 4: Проверка ping

Захват на линии PC1 — коммутатор

Запущен перехват пакетов на выбранном участке.

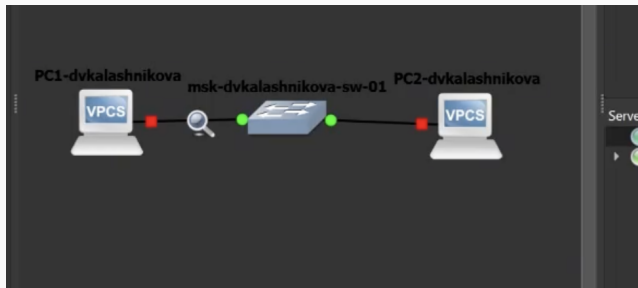
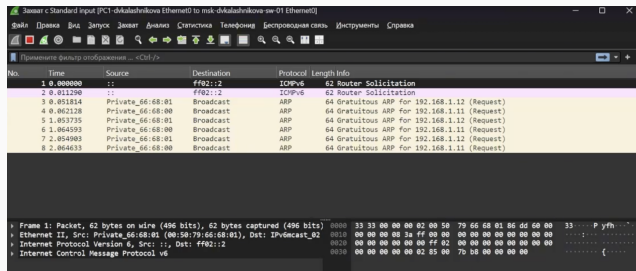


Рис. 5: Захват пакетов

Широковещательные ARP-запросы

Кадры идут на ff:ff:ff:ff:ff:ff — MAC получателя пока неизвестен.



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::2	ICMPv6	62	Router Solicitation
2	0.011290	::	ff02::2	ICMPv6	62	Router Solicitation
3	0.051814	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
4	0.062128	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1.053735	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
6	1.064593	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
7	2.054903	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
8	2.064633	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)

Frame 1: Packet, 62 bytes on wire (496 bits), 62 bytes captured (496 bits)		0000	33 33 00 00 00 02 00 50 79 66 68 01 85 dd 60 00	33 ... P yfh
Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: IPv6cast_02		0010	00 00 00 00 3a ff 00 00 00 00 00 00 00 00 00	
Internet Protocol Version 6, Src: ::, Dst: ff02::2		0020	00 00 00 00 00 00 ff 02 00 00 00 00 00 00 00	
Internet Control Message Protocol v6		0030	00 00 00 00 02 85 00 7d b8 00 00 00 00	 {

Рис. 6: ARP-запросы

Видна вложенность Ethernet II, IPv4 и ICMP.

11	392.847431	192.168.1.12	192.168.1.11	ICMP	98 Echo (ping) request	id=0xf2ac, seq=1/256, ttl=64 (reply in 12)
12	392.847431	192.168.1.11	192.168.1.12	ICMP	98 Echo (ping) reply	id=0xf2ac, seq=1/256, ttl=64 (request in 11)

Рис. 7: ICMP Echo Request

Узел узнаёт MAC владельца IPv4 и получает ответ.

13	445.576316	192.168.1.12	192.168.1.11	ECHO	98 Request
14	445.578282	192.168.1.11	192.168.1.12	ECHO	98 Response

Рис. 8: ARP Request/Reply

Запросы и ответы чередуются — ping без потерь.

15	476.317701	192.168.1.12	192.168.1.11	TCP	74	20665 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1790094662 TSecr=0 M.
16	476.318682	192.168.1.11	192.168.1.12	TCP	54	7 → 20665 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
17	476.319699	192.168.1.12	192.168.1.11	TCP	66	20665 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1790094662 TSecr=0
18	476.321666	192.168.1.12	192.168.1.11	ICMP	122	Request
19	476.321666	192.168.1.11	192.168.1.12	TCP	54	7 → 20665 [ACK] Seq=1 Ack=57 Win=2920 Len=0
20	476.326599	192.168.1.12	192.168.1.11	TCP	66	20665 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval=1790094662 T.
21	476.327575	192.168.1.11	192.168.1.12	TCP	54	7 → 20665 [ACK] Seq=1 Ack=58 Win=2920 Len=0
22	476.328554	192.168.1.11	192.168.1.12	TCP	54	7 → 20665 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
23	476.332053	192.168.1.12	192.168.1.11	TCP	66	20665 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=1790094662 TSecr=0

Рис. 9: ICMP Echo

Замена второго VPCS на FRR

Теперь проверяется и доступ PC1 к шлюзу.

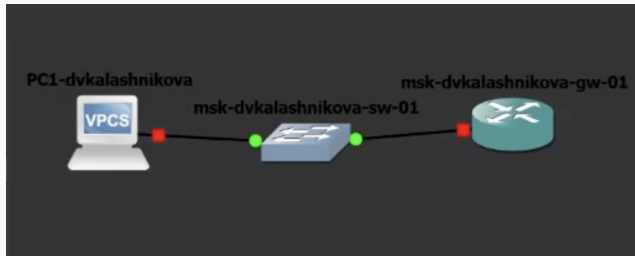
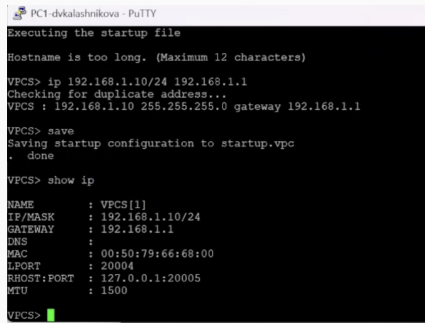


Рис. 10: FRR вместо VPCS

Повторная настройка PC1

Адрес 192.168.1.10/24, шлюз 192.168.1.1.



```
PC1-dvkalashnikova - PuTTY
Executing the startup file
Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip
NAME       : VPCS[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT : 127.0.0.1:20005
MTU        : 1500

VPCS> 
```

Рис. 11: Повторная настройка PC1

Имя msk-dvkalashnikova-gw-01, eth0 — 192.168.1.1/24.

```
Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-dvkalashnikova-gw-01
msk-dvkalashnikova-gw-01(config)# exit
msk-dvkalashnikova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dvkalashnikova-gw-01# configure terminal
msk-dvkalashnikova-gw-01(config)# interface eth0
msk-dvkalashnikova-gw-01(config-if)# ip address 192.168.1.1/24
msk-dvkalashnikova-gw-01(config-if)# no shutdown
msk-dvkalashnikova-gw-01(config-if)# exit
msk-dvkalashnikova-gw-01(config)# exit
msk-dvkalashnikova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dvkalashnikova-gw-01#
```

Рис. 12: Конфигурация FRR

Интерфейс up с адресом 192.168.1.1/24.

```
msk-dvkalashnikova-qw-01# show interface brief
Interface      Status  VRF      Addresses
-----
eth0           up      default  192.168.1.1/24
eth1           down    default
eth2           down    default
eth3           down    default
eth4           down    default
eth5           down    default
eth6           down    default
eth7           down    default
lo             up      default
pimreg         up      default
msk-dvkalashnikova-qw-01#
```

Рис. 13: Состояние eth0

Ping шлюза с PC1

Все запросы получили ответы — линия до FRR исправна.

```
VPCS> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=7.021 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=71.959 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=2.212 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=10.123 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=70.854 ms
VPCS> █
```

Рис. 14: Ping шлюза

Запросы от 192.168.1.10 и ответы от 192.168.1.1.

13	509.228332	192.168.1.10	192.168.1.1	ICMP	98 Echo (ping) request	id=0x5ab2, seq=1/256, ttl=64 (reply in 14)
14	509.235180	192.168.1.1	192.168.1.10	ICMP	98 Echo (ping) reply	id=0x5ab2, seq=1/256, ttl=64 (request in 13)
15	510.254503	192.168.1.10	192.168.1.1	ICMP	98 Echo (ping) request	id=0x5bb2, seq=2/512, ttl=64 (reply in 16)
16	510.322821	192.168.1.1	192.168.1.10	ICMP	98 Echo (ping) reply	id=0x5bb2, seq=2/512, ttl=64 (request in 15)
17	511.329911	192.168.1.10	192.168.1.1	ICMP	98 Echo (ping) request	id=0x5cb2, seq=3/768, ttl=64 (reply in 18)
18	511.331838	192.168.1.1	192.168.1.10	ICMP	98 Echo (ping) reply	id=0x5cb2, seq=3/768, ttl=64 (request in 17)
19	512.334651	192.168.1.10	192.168.1.1	ICMP	98 Echo (ping) request	id=0x5db2, seq=4/1024, ttl=64 (reply in 20)
20	512.344280	192.168.1.1	192.168.1.10	ICMP	98 Echo (ping) reply	id=0x5db2, seq=4/1024, ttl=64 (request in 19)
21	513.406882	192.168.1.10	192.168.1.1	ICMP	98 Echo (ping) request	id=0x5eb2, seq=5/1280, ttl=64 (reply in 22)
22	513.476283	192.168.1.1	192.168.1.10	ICMP	98 Echo (ping) reply	id=0x5eb2, seq=5/1280, ttl=64 (request in 21)

Рис. 15: ICMP на линии к FRR

Зелёные точки — интерфейсы подняты.

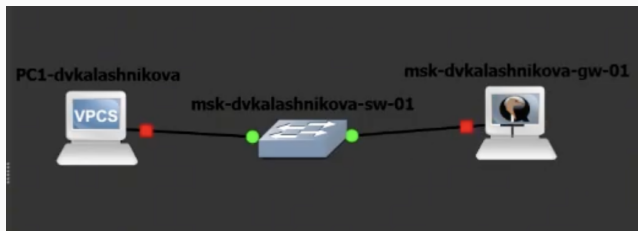


Рис. 16: Схема с FRR

Контроль show ip на PC1

Адрес 192.168.1.10/24 и шлюз 192.168.1.1 перед заменой.

```
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

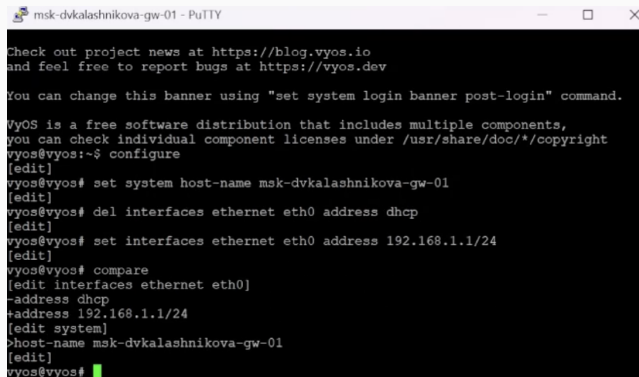
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT : 127.0.0.1:20005
MTU        : 1500

VPCS> █
```

Рис. 17: show ip на PC1

Удалён DHCP, задан статический 192.168.1.1/24 на eth0.

A screenshot of a terminal window titled "msk-dvkalashnikova-gw-01 - PuTTY". The terminal shows the VyOS configuration process. It starts with a banner message about project news and bug reports. Then, the user enters the 'configure' command. Inside the configuration mode, the user sets the host name to 'msk-dvkalashnikova-gw-01', deletes the DHCP configuration for the 'eth0' interface, and sets a static IP address of '192.168.1.1/24' for the same interface. Finally, the user enters the 'compare' command, which shows the changes: removing the DHCP address and adding the static IP. The terminal ends with the prompt 'vyos@vyos#'.

```
msk-dvkalashnikova-gw-01 - PuTTY
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

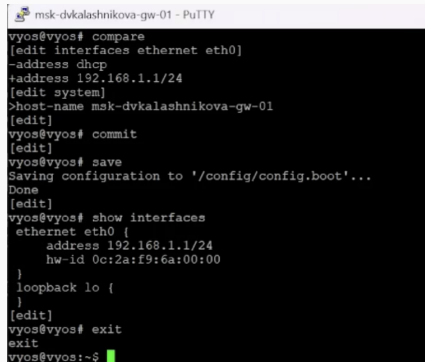
You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-dvkalashnikova-gw-01
[edit]
vyos@vyos# del interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
- address dhcp
+ address 192.168.1.1/24
[edit system]
> host-name msk-dvkalashnikova-gw-01
[edit]
vyos@vyos#
```

Рис. 18: Настройка VyOS

Проверка VyOS после commit

eth0 в состоянии up с адресом 192.168.1.1/24.



```
msk-dvkalashnikova-gw-01 - PuTTY
vyos@vyos# compare
[edit interfaces ethernet eth0]
-address dhcp
+address 192.168.1.1/24
[edit system]
>host-name msk-dvkalashnikova-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:2a:f9:6a:00:00
  }
  loopback lo {
  }
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$
```

Рис. 19: Проверка VyOS

Результат совпадает с FRR — логика IPv4 не меняется.

```
VPCS> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=35.594 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=3.825 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=30.853 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=33.247 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=2.646 ms
VPCS> █
```

Рис. 20: Ping через VyOS

Wireshark показывает запросы и ответы между 192.168.1.10 и 192.168.1.1.

85	3526.455166	192.168.1.10	192.168.1.1	ICMP	98 Echo (ping) request	id=0xfac1, seq=1/256, ttl=64 (reply in 86)
86	3526.490273	192.168.1.1	192.168.1.10	ICMP	98 Echo (ping) reply	id=0xfac1, seq=1/256, ttl=64 (request in 85)
87	3527.518321	192.168.1.10	192.168.1.1	ICMP	98 Echo (ping) request	id=0xfbc1, seq=2/512, ttl=64 (reply in 88)
88	3527.521427	192.168.1.1	192.168.1.10	ICMP	98 Echo (ping) reply	id=0xfbc1, seq=2/512, ttl=64 (request in 87)
89	3528.610866	192.168.1.10	192.168.1.1	ICMP	98 Echo (ping) request	id=0xfdc1, seq=3/768, ttl=64 (reply in 90)
90	3528.613474	192.168.1.1	192.168.1.10	ICMP	98 Echo (ping) reply	id=0xfdc1, seq=3/768, ttl=64 (request in 89)
91	3529.678099	192.168.1.10	192.168.1.1	ICMP	98 Echo (ping) request	id=0xfec1, seq=4/1024, ttl=64 (reply in 92)
92	3529.683560	192.168.1.1	192.168.1.10	ICMP	98 Echo (ping) reply	id=0xfec1, seq=4/1024, ttl=64 (request in 91)
93	3530.735298	192.168.1.10	192.168.1.1	ICMP	98 Echo (ping) request	id=0xffc1, seq=5/1280, ttl=64 (reply in 94)
94	3530.737433	192.168.1.1	192.168.1.10	ICMP	98 Echo (ping) reply	id=0xffc1, seq=5/1280, ttl=64 (request in 93)

Рис. 21: ICMP с VyOS

Освоено построение простых схем на коммутаторе и маршрутизаторе. Изучено поведение Ethernet, ARP и ICMP в Wireshark. Стенд работает корректно.